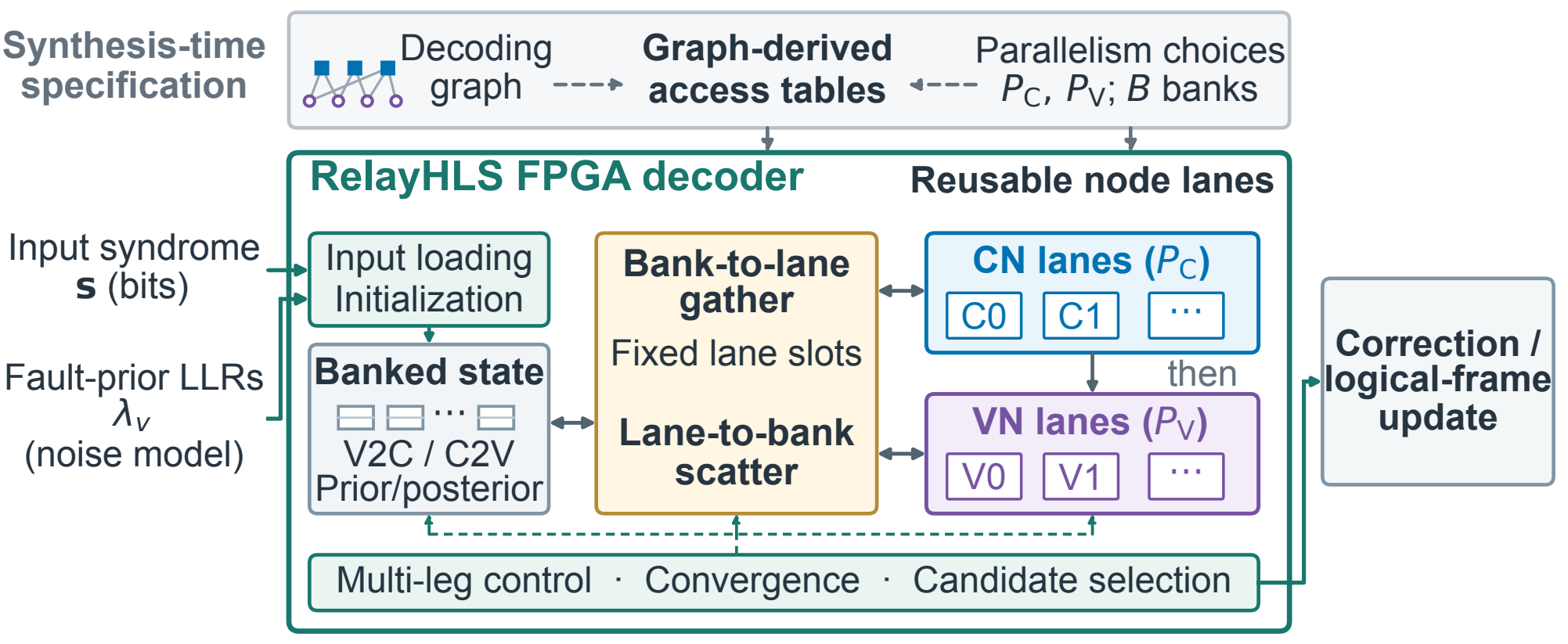

Synthesis-time specification
Decoding graph
Graph-derived access tables
Parallelism choices
P_C, P_V; B banks
RelayHLS FPGA decoder
Reusable node lanes
Input syndrome
s (bits)
Fault-prior LLRs
λ_v
(noise model)
Input loading
Initialization
Banked state
V2C / C2V
Prior/posterior
Bank-to-lane gather
Fixed lane slots
Lane-to-bank scatter
CN lanes (P_C)
C0
C1
⋯
then
VN lanes (P_V)
V0
V1
⋯
Multi-leg control · Convergence · Candidate selection
Correction / logical-frame update